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Garden of Knowledge and Virtue

TITLE

Smart Object Finder: A Safety and Mobility Enhancement Tool for the Visually Impaired

AUTHORS

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ABSTRACT

This paper presents the development of a Smart Object Finder system designed to enhance safety and independence for elderly and visually impaired users through intelligent object detection and user-friendly feedback. The proposed system integrates a vision-based approach using the HuskyLens AI camera together with a microcontroller-based control unit to identify and locate commonly used objects in the surrounding environment. The system is divided into two main subsystems: a user input and communication system, and a vision-based object detection system. Users interact with the cane through push buttons to select predefined target objects, while audio feedback in the form of buzzer signals provides real-time guidance based on the detected object's distance and position. By combining embedded systems, wireless communication, and artificial intelligence vision capabilities, this project demonstrates an effective mechatronics system integration that improves situational awareness and navigation assistance. Experimental testing confirms that the system is responsive, reliable, and suitable for assisting visually impaired users in locating objects within indoor environments.

1. INTRODUCTION

Visual impairment and age-related mobility limitations pose significant challenges to daily living, particularly in terms of navigation, object recognition, and personal safety. Elderly and visually impaired individuals often rely on conventional canes as primary assistive tools; however, these traditional aids are limited to ground-level obstacle detection and do not provide information about surrounding objects or contextual awareness. As a result, users may experience difficulties locating essential items, navigating unfamiliar environments, and responding effectively to potentially hazardous situations.

Advancements in assistive technology and embedded systems have enabled the transformation of conventional mobility aids into intelligent devices that enhance user independence and safety. Previous studies have shown that integrating sensors, audio or haptic

feedback, and microcontroller-based processing can significantly improve environmental awareness for visually impaired users. More recent developments have incorporated artificial intelligence (AI) and computer vision to allow assistive devices to recognize objects and provide more meaningful feedback, moving beyond simple distance-based detection.

In line with these developments, this project proposes a Smart Cane system that utilizes a HuskyLens AI vision sensor for object detection and recognition. The system is designed based on a modular mechatronics system integration (MSI) approach, where processing tasks are distributed between hardware components to ensure efficient and reliable operation. The cane allows users to select specific target objects via push buttons, while the vision system detects the selected object and communicates its position to generate audio feedback. This approach enables visually impaired users to locate common objects more intuitively without relying solely on physical contact.

The main objective of this project is to design, implement, and evaluate a Smart Cane that enhances object-finding capability and user interaction through vision-based detection and embedded system integration. By combining AI vision, wireless communication, and real-time feedback, the proposed system aims to demonstrate a practical and low-cost assistive solution that supports safer and more independent daily activities for elderly and visually impaired individuals.

Table 1 Table of comparison

Author / System	Main Technology Used	Detection Capability	User Feedback Method	Limitations
Dey et al. [9]	Ultrasonic sensor, PIC microcontroller	Obstacle detection (short range)	Buzzer / vibration	Cannot recognize specific objects, limited contextual awareness
Gbenga [10]	Ultrasonic sensor, PIC microcontroller	Obstacle detection (up to 3 m)	Audio alert	No intelligent object

				recognition, no user selection
IoT-based Smart Cane [11]	Ultrasonic sensor, GPS, GSM	Obstacle detection and location tracking	Vibration, buzzer, LED	High power consumption, no vision-based detection
IR & ToF-based Cane [12][13]	IR camera, ToF sensors	Obstacle and water detection	Vibration and audio	Limited object classification, complex hardware
IR & ToF-based Cane [12][13]	IR camera, ToF sensors	Obstacle and water detection	Vibration and audio	Limited object classification, complex hardware
Mobile App–Integrate	Smartphone camera, AI algorithms	Object and environment recognition	Audio via mobile device	Dependence on smartphone, less portable
Proposed Smart Object Finder (This Project)	HuskyLens AI camera, ESP32, Bluetooth	Vision-based object detection and localization	Audio feedback (buzzer)	Limited to pre-trained objects, indoor-focused testing

2. METHODOLOGY

2.1 Project Development

Overall, a block diagram is used to illustrate the complete operation of the Smart Object Finder for the Visually Impaired system, showing the relationship between the input, processing, communication and output components. The project is divided into three main systems, namely System 1 (User Input and Communication System) and System 2 (Vision-Based Object Detection System). Fig. 1 shows the block diagram of System 1, Fig. 2 presents the block diagram of System 2, while Fig. 3 illustrates the flowchart that explains the working principle of the entire system.

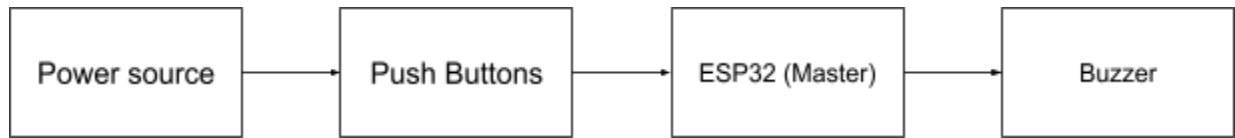


Fig. 1 Block diagram of system 1 User Input and Communication System

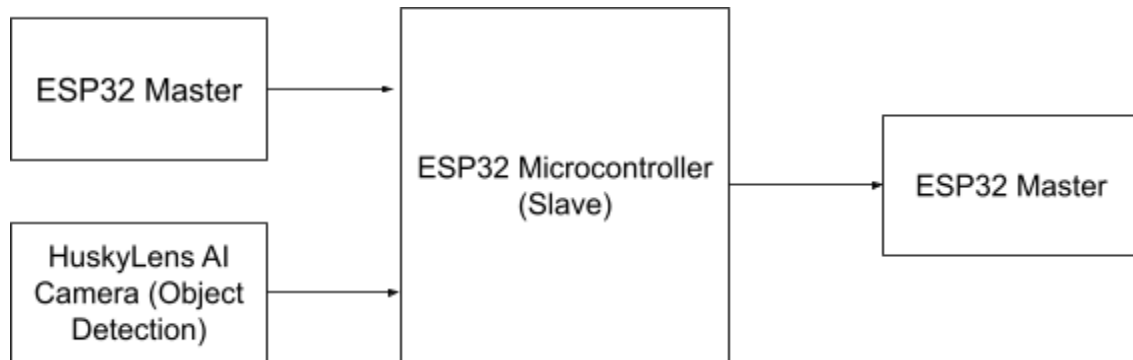


Fig. 2 Block diagram of system 2 Vision-Based Object Detection System

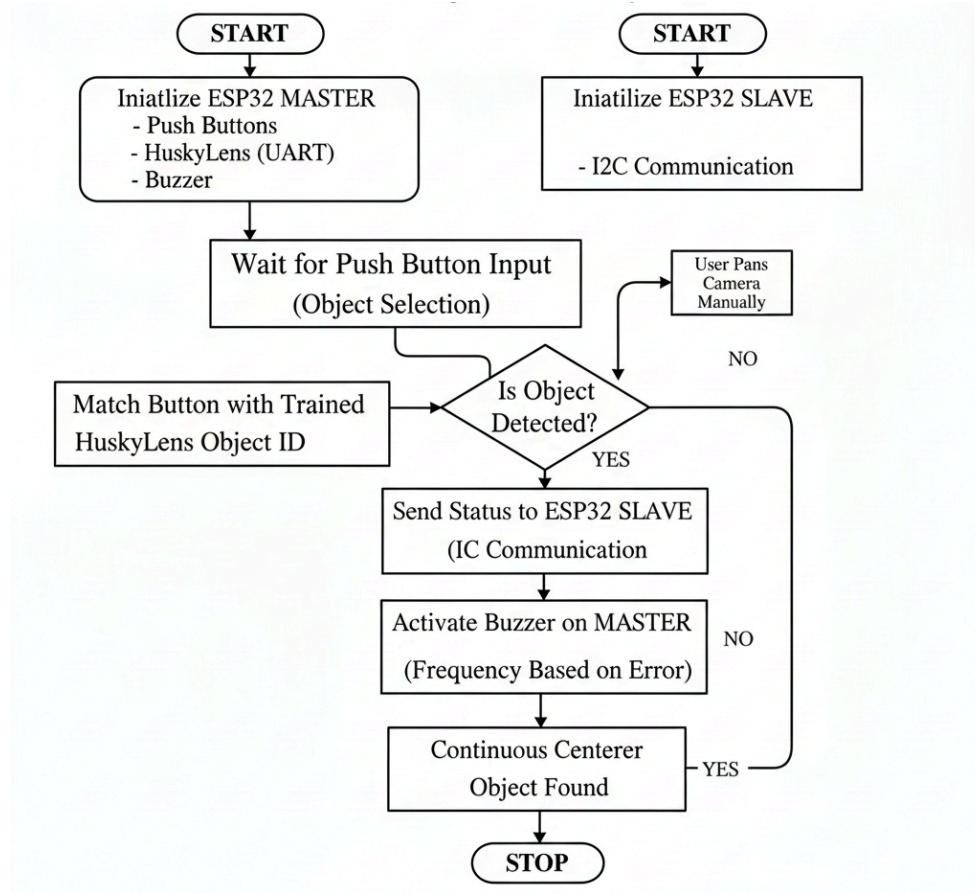


Fig. 3 The flow chart that explains all the work principles for the Smart Finder Object project.

2.2 Schematic Diagram and Prototype Setup

Figure 2.2.1 illustrates the circuit diagram of System 1, which serves as the user interface and feedback unit of the proposed assistive system for the visually impaired. This system consists of an ESP32 microcontroller interfaced with three push buttons that represent different lost items, allowing the user to select the desired object. Additionally, a buzzer module is integrated to provide audio feedback based on the distance and orientation of the detected object, thereby guiding the user during the searching process.

Meanwhile, Figure 2.2.2 presents the circuit diagram of System 2, which functions as the vision and object detection unit. This system comprises an ESP32 microcontroller integrated with a HuskyLens AI vision sensor, where the target objects are pre-trained and stored. Upon receiving the selected item information from System 1 via Bluetooth communication, the HuskyLens detects the corresponding object and outputs its positional data, which is processed and transmitted back to System 1 to generate appropriate audio feedback for the user.

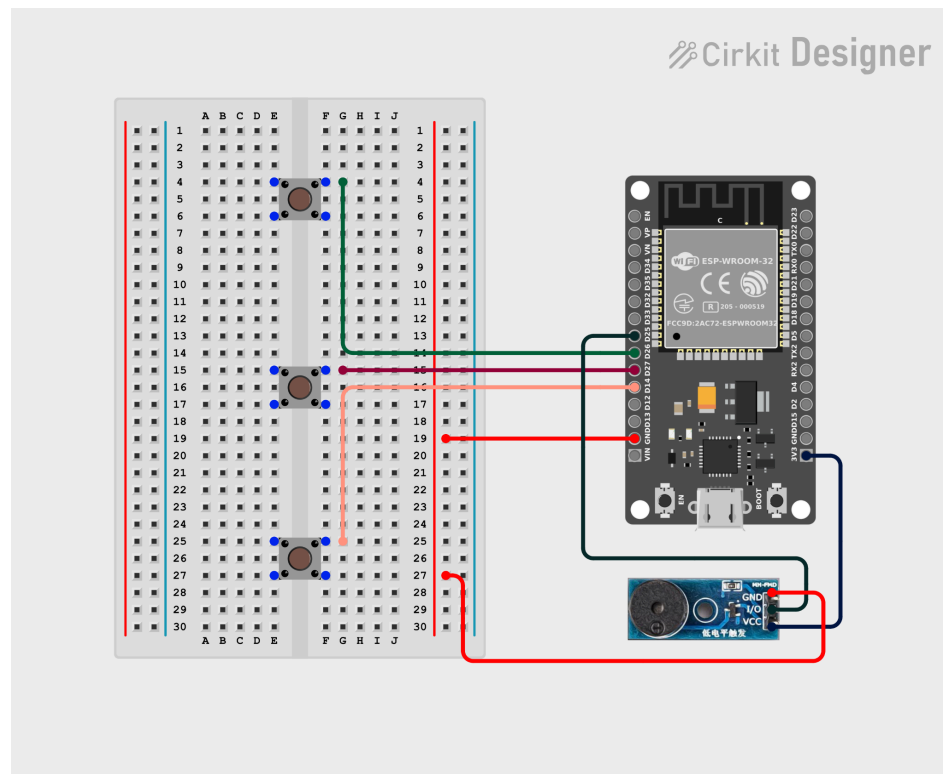


Figure 2.2.1 Electrical Design of System 1

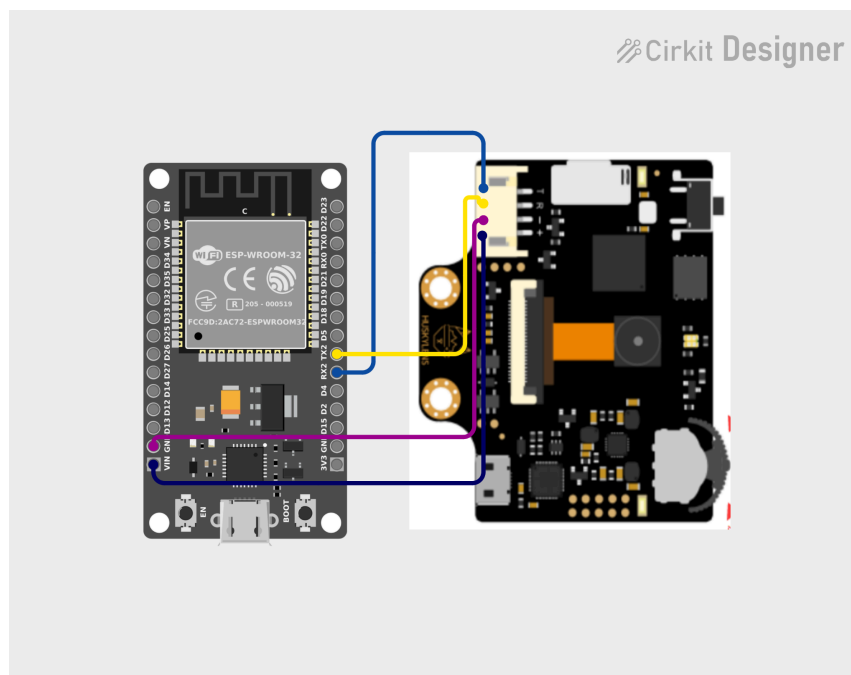


Figure 2.2.2 Electrical Design of System 2

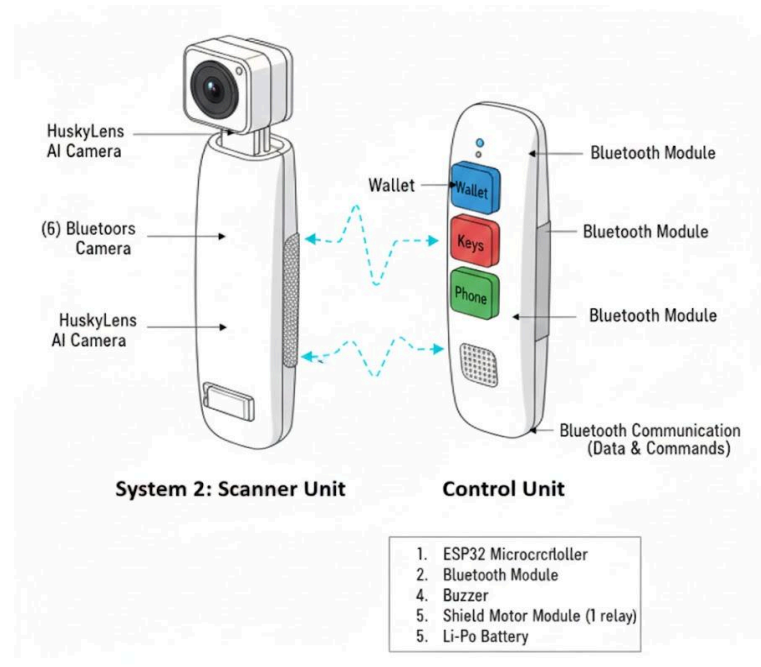


Figure 2.2.3 Prototype Setup (Gemini)